Review

SCIENCE PROGRESS

Science Progress
2025, Vol. 108(4) 1–33
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DOI: 10.1177/00368504251385937
journals.sagepub.com/home/sci

The role of genetic and epigenetic factors in sports-related muscle, bone, and brain injuries

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Abstract

Understanding how genetic and epigenetic factors influence predisposition, susceptibility, protection, severity, and recovery potential in sports-related muscle, bone, and brain injuries is gaining increasing attention, as these injuries represent a major concern for athletes' health and performance. This narrative review explores the role of specific genetic variants, such as those in *ACE, ACTN3, AMPD1, VDR*, and *APOE*, in predisposing athletes to injury. In parallel, the emerging role of microRNAs, including miR-21, miR-133a, miR-208a, and miR-214 in regulating gene expression related to muscle regeneration, inflammation, and neuroprotection, is highlighted. These small noncoding RNAs are influenced by environmental stimuli such as sports activity and training, and may serve as promising biomarkers

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for injury risk and recovery. This review is based on a narrative approach, involving targeted literature searches in databases such as PubMed, Scopus, and Web of Science, with no restrictions on language or publication date. Articles were selected based on relevance using predefined keywords and were categorized by injury type (muscular, bone, and concussion-related). However, several limitations remain, including population heterogeneity, small sample sizes, and the lack of functional validation of identified variants, which currently limits their clinical application. Ethical considerations surrounding the use of genetic data in athletes also need to be addressed. In the future, better integration of multiomic data could support the development of personalized and ethically sound strategies for injury prevention and athlete care.

Keywords
Concussions, epigenetics, musculoskeletal injuries, SNP, sports genetics

Introduction

The predisposition to injury and sports performance is primarily determined by intrinsic biological traits, including genetic polymorphisms that influence muscle structure, bone density, and neuromuscular function. While training and other extrinsic factors can modulate injury risk, they do not define the baseline predisposition, which is rooted in these intrinsic characteristics.

The Healthy Injury-Free Adolescent Athletes study has shown that multiple factors, including biomechanical, environmental, and psychosocial aspects, contribute significantly to injury risk in young athletes.[1] Similarly, studies on Olympic athletes have demonstrated that elite-level performance tends to run in families, emphasizing the important contribution of genetic background.[2] Foundational studies further suggest that up to 60–70% of the variability in physiological adaptations to physical activity (such as VO2max and strength gains) and injury susceptibility may be explained by genetic factors.[3] Engaging in intense physical activity comes with an increased risk of injuries, the nature and severity of which vary depending on the sport. Indeed, each sport has a specific risk profile, related to its unique physical and technical demands. Nearly 70% of track and field athletes experience injuries per season.[4] Lower limb injuries are extremely common in certain sports, affecting 50% of athletes participating in football, marathon, triathlon, gymnastics (female), basketball, pole vaulting, rugby sevens, volleyball (female), handball, and tennis.[5] Among these injuries, muscular injuries such as hamstring strains are particularly prevalent; for instance, professional footballers experience a 20% hamstring injury rate per season, with 20% of these progressing to a chronic stage.[6] Stress fractures and stress reactions, while less frequent, pose a greater challenge due to their longer recovery periods, with athletes requiring on average 67.4 days (± 73.1) to return to full training for stress reactions, and up to 199 days (± 205.2) for stress fractures.[7] In comparison, concussions, though less common, are a growing concern in contact sports, with rugby union players experiencing concussion rates between 8.2 and 16.11 per 1000 h of activity, Gaelic football players 5.21 per 1000 h, and football players

between 2.08 and 4.04 per 1000 activity events.[8] Beyond the biological and environmental factors influencing predisposition and injury risk, it is important to consider the significant economic impact of sports injuries. A recent literature review revealed that, despite methodological heterogeneity, studies consistently report substantial direct and indirect costs associated with sports injuries, including medical expenses, income loss, and reduced quality of life.[9] For example, a study conducted in Switzerland analyzing 702 injuries in amateur football estimated an average cost of €4030 per injury, with costs reaching up to €8190 for players over 30 competing in specific amateur leagues.[10] These figures illustrate the considerable burden sports injuries place on healthcare systems and individuals, underscoring the urgent need for effective prevention and management strategies tailored to the specific demands of each sport.

Research has increasingly focused on the genetic basis of athletic status. For example, a genome-wide linkage scan involving 700 pairs of British dizygotic twins estimated that approximately 66% of the variance in athlete status could be explained by genetic factors, according to the specific assumptions of the twin study design. This estimate, however, varies across sport disciplines and depends on the criteria used to define athlete status.[11] Several studies have also reported associations between specific genetic polymorphisms and the predisposition to distinct categories of sports-related injuries, such as those affecting muscle tissue, bone structure, and concussions, which differ substantially in their underlying pathophysiology.[1,12–16] Genetic susceptibility is most often investigated through the analysis of single-nucleotide polymorphisms (SNPs), the most prevalent form of genetic variation in the human genome. Other types of variants, including insertion–deletion polymorphisms, have likewise been implicated in modulating injury risk and recovery, potentially through effects on gene expression or protein function. Each SNP may give rise to distinct genotypes, such as homozygous (e.g., AA or TT) or heterozygous (e.g., AT) combinations. These genotypes can influence gene expression or protein function and are frequently examined to investigate associations with athletic performance, injury risk, or recovery potential.[17]

Epigenetic mechanisms, such as DNA methylation and microRNAs (miRNAs), modulate gene activity and are influenced by exercise and environmental factors. MicroRNAs are a class of small, noncoding RNA molecules approximately 20–22 nucleotides long that regulate gene expression at the posttranscriptional level by binding to messenger RNAs and promoting their degradation or inhibiting their translation. They are key regulators of biological processes, including tissue repair, inflammation, and muscle regeneration, thereby impacting injury susceptibility and recovery dynamics. In the context of concussions, they are involved in neuroinflammatory responses and neuronal repair. Understanding these genetic and epigenetic contributions enables sports medicine professionals to identify athletes at higher risk of injury, to personalize training and recovery protocols based on each athlete's genetic and epigenetic profile, which may influence muscle repair capacity, inflammatory responses, bone remodeling, and neurological recovery. This could involve aligning training variables, such as intensity, type (e.g., explosive vs. endurance emphasis), volume, and recovery duration, with the athlete's specific biological predispositions. For example, individuals with SNPs or miRNA expression patterns linked to slower tissue repair or higher inflammation might benefit from more conservative return-to-play timelines, targeted anti-inflammatory strategies, longer rest

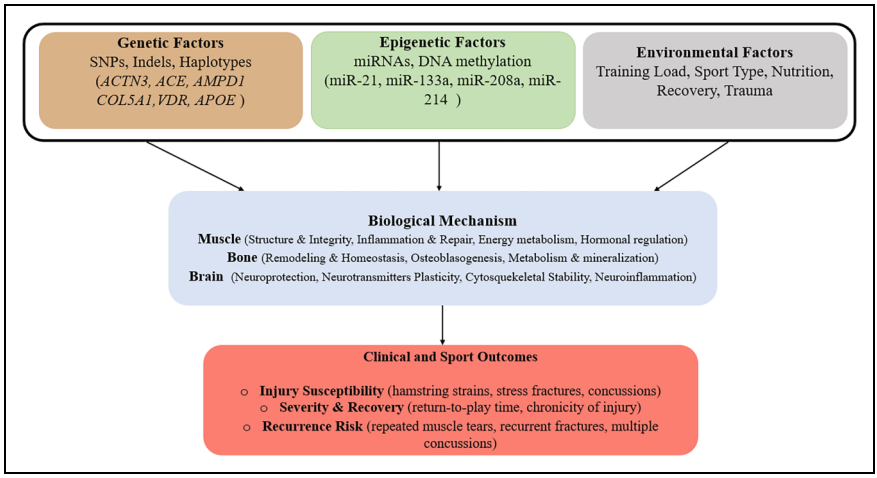

Figure 1. Conceptual framework illustrating how genetic polymorphisms and epigenetic regulators (e.g., microRNAs) interact with muscle, bone, and brain biological pathways, while being modulated by environmental factors (training load, sport type, nutrition, trauma). These interactions collectively influence injury susceptibility, severity, recovery, and recurrence risk.

intervals, or tailored nutritional support. Conversely, athletes with profiles indicating faster recovery potential could sustain higher training volumes or shorter recovery periods without increasing injury risk.[18–23] A conceptual framework summarizing the interplay between genetic, epigenetic, and environmental factors influencing sports-related injuries is presented in Figure 1.

To our knowledge, this narrative review is the first to integrate genetic and epigenetic evidence related to muscle, bone, and concussion injuries in athletes. It consolidates current findings, identifies the most frequently implicated genes and biological mechanisms, and discusses their potential applications for personalized injury prevention strategies in sports medicine.

Methodologie

A nonsystematic search of the scientific literature was conducted in PubMed, Scopus, Web of Science, EBSCO, and Google Scholar, covering publications up to 2025. Keywords included: “sports injuries”, “muscle injuries”, “stress fractures”, “concussions”, “genetic polymorphisms”, “sports genetics”, and “miRNA”, combined with “athletes”, “susceptibility”, and “epigenetics”. Additional searches used related terms and synonyms (e.g., “soft tissue injuries”, “bone stress injury”, “traumatic brain injury”,"gene variants”, “non-coding RNA”) to maximize coverage. Eligible publications included original research articles, narrative or systematic reviews, focused on human studies involving athletes or physically active individuals. Since this is not a systematic review, article selection was based on relevance to the topic, scientific value, and contribution to

Table 1. Genetic variants associated with muscle damage in athletes.

Genes	Full name	Polymorphisms	Localization	References
ACE	Angiotensin-converting enzyme	rs1799752	17q23.3	28,29–31,32
ACTN3	Actinin alpha 3	rs1815739	11q13.2	28,33–36
AMPD1	Adenosine monophosphate deaminase 1	rs17602729	1p13.2	15,16,33
CCR2	C-C chemokine receptor type 2	rs3918358 rs768539	3p21.31	37
CKM	Creatine kinase M-type	rs8111989	19q13.32	16,33
CNTF	Ciliary neurotrophic factor	rs2515362	11q12.1	38
COL3A1	Collagen type III alpha 1 chain	rs1800255	2q32.2	39
COL5A1	Collagen type V alpha 1 chain	rs12722 rs3196378	9q34.3	27,29,40,41
ESR1	Estrogen receptor 1	rs2234693	6q25.1-q25.2	42
GEFT (ARHGEF25)	Guanine nucleotide exchange factor	rs11613457	12q13.3	43
HGF	Hepatocyte growth factor	rs5745678 rs5745697	7q21.11	43
LIF	Adrenoceptor beta 2	rs9290271	22q12.2	43
MCT1	Monocarboxylate transporter 1	rs1049434	1p13.2	29,44,45
MLCK/ MYLK	Myosin light chain kinase	rs28497577 rs2700352	3q21.1	16,33,46
NOG	Noggin	rs1372857	17q22	40
NOS3	Nitric oxide synthase	rs1799983	7q36.1	47
SOX15	SRY-box transcription factor 15	rs4227	17p13.1	43

understanding injury mechanisms. To complement the database search, reference lists of selected articles were screened manually, and citation tracking was used to identify further relevant studies.

Genetic polymorphisms related to muscle injuries in athletes

Muscle injuries result from damage to muscle fibers, caused by, for example, a contusion (sudden impact) or a strain (excessive tensile force). These injuries often lead to myofiber rupture, typically near the myotendinous junction.[24] Common in sports, they can be classified by mechanism, location, severity, and recurrence frequency.[25] Research has shown associations between various genetic variants and susceptibility to muscle injuries in athletes (Table 1), focusing on genes involved in muscle structure, regeneration, energy metabolism, contraction, and the regulation of inflammation and oxidative stress response.[26,27] This review identified 17 genes and 21 polymorphisms linked to muscle injuries.

Genes, muscle structure, and integrity in athletes

Intensive exercise can induce changes in muscle fiber structure and organization, which are influenced by several genes associated with muscle composition, function, and

particularly muscle contraction. These genetic variants enable athletes to adapt better to training while resisting muscle injuries. Collagen plays a vital role in regulating collagen fibril formation and organization in various connective tissues, including skeletal muscle.[27] Genetic variants influencing muscle structure and integrity are crucial for maintaining the stability and elongation of muscle fibers. One significant variant is the R577X or C577 T polymorphism (rs1815739) in the *ACTN3* gene, where arginine (R) is replaced by a stop codon (X) at position 577. This polymorphism is also referred to using the nucleotide nomenclature, with R = C and X = T. Individuals with the homozygous XX genotype lack α-actinin-3, making them more prone to muscle injuries.[29] Whereas those with the RR or RX genotypes produce α-actinin-3, with higher expression in RR individuals.[48] According to the literature, results on *ACTN3* are not uniform across studies. Six studies reported no association between the *ACTN3* polymorphism and muscle injuries, particularly in football and rugby players,[16,49–51] as well as in runners and endurance athletes.[48,52] However, a larger group of studies has reported associations between R577X polymorphism of the *ACTN3* gene and an increased predisposition to muscle injuries in football players,[26,28,29,33,35,50,53,54] half marathon runners,[55] and marathon runners.[56] This vulnerability is attributed to increased instability of the Z discs and, consequently, the sarcomere, resulting in excessive polymerization of actin filaments. This disruption can compromise the effective transmission of muscle contraction signals and lead to a decrease in the force generated by muscle fibers.[53]

Among the studies that found this association, the X allele is associated with an increased risk of muscle injuries compared to the R allele.[28,33,34] Moreover, the XX genotype appears linked to longer recovery times[35] and lower match performance values than RR players..[53] In contrast, the R allele is significantly more prevalent among female athletes with muscle injuries in sports such as football, softball, basketball, and badminton ($p = 0.04$),[36] and also in those with injuries related to endurance running training.[56] The recent narrative review by Kahya and Taheri, which provides a stratified synthesis of genetic markers by sport type and gene function, supports the association of *ACTN3* R577X polymorphism with soft-tissue injury risk, particularly in strength- and endurance-based disciplines.[57] Additionally, the R allele of *ACTN3* has been shown to protect football players from developing muscle injuries[29] and is linked to a decrease in serum creatine kinase (CK) levels.[55] Pimenta et al. showed that CK levels (4 h after exercise), cortisol levels (2 h after exercise), and alpha-actin levels (postexercise) were generally higher in Brazilian professional football players with the *ACTN3* XX genotype. However, the results also showed that the R allele was significantly associated with higher levels of interleukin-6 (IL-6).[54] Finally, a marginal association was detected for *ACTN3* rs1815739, where athletes with the R allele seem more likely to experience muscle pain (MP) after exercise than those with the XX genotype.[58] A systematic review found that the XX genotype is associated with sports injuries, particularly among elite athletes.[59] Overall, *ACTN3* polymorphisms appear to influence injury risk through both structural mechanisms (Z-disc and sarcomere instability) and biochemical pathways (CK, cortisol, IL-6). These effects are modulated by athlete sex, sport type, and recovery response. This integrative view reconciles divergent findings across studies and suggests that *ACTN3* could serve as a practical marker to guide personalized training loads and recovery strategies.

Several studies have highlighted the role of the *MLCK* gene in susceptibility to muscle injuries, with two polymorphisms of the *MLCK* gene linked to a decrease in postexercise strength. Heterozygous CA individuals for the c.37885 C > A polymorphism may experience greater muscle damage after endurance competitions compared to those with the CC genotype.[46] and the frequency of the CA genotype among injured athletes (32.7%) was not higher than that observed in uninjured athletes (48.9%).[52] In contrast, this polymorphism is associated with an increased risk of muscle tears in professional football players, where the CA genotype is linked to a higher incidence of match injuries, while the CC genotype offers protection.[16] Furthermore, the GT genotype of the *MYLK* gene rs28497577 was associated with prolonged absence due to a knee injury in football players.[60] Moreover, polymorphisms in the *MLCK* gene were significantly associated with injuries in amateur female football players, with a different genotypic distribution between injured and uninjured players ($p = 0.031$).[33] However, CC genotype for the *MLCK* gene polymorphism c.37885 C > A exhibited higher serum myoglobin concentrations and greater pre-to-post-race leg muscle power reduction compared to those with the CA genotype,[61] and also the AA genotype was found to be advantageous for increasing body mass index (BMI) after creatine supplementation compared to the CA and CC genotypes.[49] These findings suggest that the impact of *MLCK* variants is context-dependent: in some cases (e.g., football players), the CA genotype increases injury susceptibility, while in others (e.g., endurance runners), it is associated with lower risk but greater postexercise muscle damage. This dual role highlights the importance of sport-specific demands and recovery dynamics and suggests that *MLCK* polymorphisms could guide personalized strengthening programs and postexercise monitoring.

In soccer players, research has shown that the GA genotype (c.*800A > G) of the *CKM* gene had a higher frequency of musculoskeletal injuries and increased injury severity during the season. Furthermore, players with the GG genotype have a higher incidence of injuries during training compared to matches.[16] The GG genotype had a higher frequency in training injuries compared to those observed in matches ($p = 0.038$).[33] Additionally, in athletes, the frequency of the A allele is significantly higher among those who have sustained injuries (47.3%) compared to those who have not (42.2%).[52] Players with the GG genotype (c.*800A > G) also demonstrate an increased response in terms of muscle mass and BMI during creatine supplementation, highlighting the importance of this genotype in the performance of power athletes.[49] Overall, *CKM* polymorphisms appear to influence both injury risk and recovery potential, while also modulating responses to supplementation, suggesting their value as markers for adjusting training load and optimizing performance in strength-oriented disciplines. Furthermore, the rs1800255 polymorphism of the *COL3A1* gene has been examined in two studies has examined the rs1800255 polymorphism of the *COL3A1* gene in relation to muscle injuries in athletes. The first study found an association between the A allele of this polymorphism and rugby athlete status and increased resistance to soft tissue injuries.[39] In contrast, the second study found no correlation between this polymorphism and exercise-induced muscle cramps.[62] Although findings remain inconsistent, evidence suggests that *COL3A1* variants may influence soft tissue resilience, which could be considered in training planning and injury prevention strategies. Indeed, the rs12722 polymorphism of the *COL5A1* gene is associated with muscle injuries in elite Australian Football League players.[40] Additionally, the C allele of *COL5A1* has

been associated with a reduced risk of muscle injuries in Italian football players.[29] Similarly, another study revealed that *COL5A1* rs12722 accounted for 44% of injury severity ($p = 0.002$).[41] These results are consistent with those of Heffernan et al., who reported a significant association between the inferred C-C allele combination, meaning the simultaneous presence of the C allele at both the rs12722 and rs3196378 polymorphic sites, and protection against muscle injuries.[27] Similarly, O'Connell et al., found that the CC genotype was associated with protection against exercise-associated muscle cramping.[62] However, two other studies found no association between the different rs12722 genotypes and muscle injury risk.[63,64] However, other studies failed to detect associations between rs12722 genotypes and injury risk.[63,64] These discrepancies suggest that *COL5A1* may influence injury susceptibility in a sport- or context-dependent manner.

Genes, inflammatory response, and muscle repair mechanisms in athletes

The inflammatory phase is the first stage of muscle repair, characterized by the disorganization of myofibrils and vascular ruptures following mechanical or metabolic stress.[65] This structural disruption leads to the release of pro-inflammatory mediators, such as cytokines and chemokines, into the extracellular space. These signals attract immune cells to the site of injury and contribute to the formation of a hematoma.[66] This process also activates proteolytic enzymes, stimulates cell proliferation and migration, and initiates the repair of damaged cell membranes. In addition, calcium influx triggers several intracellular signaling pathways, such as NF-κB, MAPK, PI3 K/Akt, and JAK/STAT, that facilitate the removal of cellular debris and promote tissue regeneration.[67] In this context, genetic polymorphisms may influence the inflammatory response and the extent of muscle recovery. For example, the *CCR2* gene variants rs768539 and rs3918358 have been associated with reduced pre-exercise strength, impaired strength recovery, and increased pain perception in some individuals.[37] Furthermore, no association was found between the rs768539 polymorphism of the *CCR2* gene and hamstring injuries in football players.[47] *CCR2* polymorphisms appear to modulate the intensity of the inflammatory response and the efficiency of muscle recovery, although their impact on actual injury risk remains uncertain. Similarly, the GG genotype of the *CNTF* gene (rs2515362) has been associated with better adaptability to strength training, potentially leading to greater muscle gains, improved resistance to injuries, and enhanced fatigue tolerance in sports school students.[38] This suggests that *CNTF* variants may act as modulators of neuromuscular adaptation, with implications for both performance and resilience to injury. In addition, a study conducted among football players revealed a significant association between the rs9290271 polymorphism of the *LIF* gene and muscle injuries and recovery time.[43] These findings indicate that *LIF* polymorphisms could influence tissue repair dynamics, potentially serving as markers for predicting recovery duration after muscle injuries. Finally, the T allele of the *HGF* gene (rs5745678) seems to offer protection against severe injuries, likely by enhancing *HGF*-mediated muscle regeneration processes such as activation of muscle stem cells, promotion of anti-inflammatory macrophage phenotypes, and inhibition of fibrosis, which collectively contribute to faster recovery times.[68] Furthermore, the CA/AA genotypes (rs5745697) are linked to minor

injuries and also promote faster recovery in football players.[43] Overall, *HGF* polymorphisms appear to support regenerative capacity and recovery quality, reinforcing their potential importance in injury prevention strategies.

Genes and muscle energy metabolism in athletes

Energetic regulation plays an important role in maintaining cellular energy homeostasis, thus ensuring better muscle contraction function and movement. The accumulation of waste products can damage cells and make them more susceptible to injuries during high-intensity activities, which is particularly noticeable in certain athletes. This is due to disruptions caused by genetic variants that affect energy metabolism, as well as oxidative stress related to waste accumulation and a decrease in the transport of metabolites and energy substrates.[15,16,33] The polymorphism c.34C > T (rs17602729) of the *AMPD1* gene has been extensively studied in the context of sports injuries. Research has shown that individuals carrying the T allele, especially those with the TT genotype, are more likely to develop metabolic myopathies characterized by exercise-induced muscle symptoms such as early fatigue, cramps, and myalgias. These symptoms increase their injury risk, particularly in demanding sports like football.[15,16,33] Furthermore, athletes with the TT genotype exhibit decreased exercise capacity and cardiorespiratory responses, as well as a limited response to endurance training, which may also contribute to an increased risk of injury. Studies indicate that football players with the CC genotype of the *AMPD1* gene experience fewer ankle and knee injuries and play more matches than those expressing a T allele. This is true for both professional and amateur football players, suggesting that the CC genotype may offer protection against injuries.[15,16,33] This link between the *AMPD1* polymorphism and sports injuries is particularly relevant for endurance athletes, where optimal muscle metabolism is essential for injury prevention.[52] Overall, *AMPD1* polymorphisms appear to shape individual susceptibility to muscle injuries by influencing energy metabolism efficiency and tolerance to high physical demands. Moreover, the AA genotype of the gene *MCT1* (rs1049434) is associated with a significant increase in muscle injuries in elite football players compared to the TT genotype.[44] Furthermore, Italian football players with the TT genotype have a significantly higher percentage of lean body mass than those with the AA genotype, suggesting that the T allele is linked to better body composition in young athletes.[45] Additionally, another study conducted on Italian football players revealed that carriers of the T allele have a protective effect against muscle injuries.[29] One study reported a transmission distortion, meaning in this case a deviation from the Hardy–Weinberg equilibrium at the population level, of the *MCT1* rs1049434 polymorphism, with an overrepresentation of the T allele among elite Polish athletes, including both endurance athletes (3000 m, marathon, cross-country skiing, swimming, triathlon, and rowing) and power/speed athletes (100–400 m sprint, weightlifting, and powerlifting).[69] However, no significant difference was observed in the allele distribution of the A/T polymorphism (rs1049434) of the *MCT1* gene between athletes and the control group, suggesting that this polymorphism may not influence muscle injury susceptibility in young athletes preparing for the Turkish Youth Championship.[70] Taken together, *MCT1* polymorphisms may contribute

to interindividual variability in injury susceptibility and athletic performance, although findings remain population- and sport-specific.

Genes, cardiovascular, and muscle regulation in athletes

Cardiovascular and muscular regulation is essential for maintaining muscle integrity by regulating oxygen supply and blood transport through arteries and veins. Genetic variants have been associated with vasodilation issues, which can lead to a decrease in blood pressure. The imbalance in this regulation causes muscle fatigue and the accumulation of metabolic waste. This makes muscles more susceptible to injuries, especially during training. Genetic variants play a key role in cardiovascular and muscular regulation.[71,72] For example, motor activity induces dynamic changes in *ACE* gene expression in rats, with bioinformatics analysis linking these shifts to cardiovascular regulation and electrolyte balance during endurance exercise.[73] A common genetic variation known as the *ACE I/D* polymorphism (rs1799752) alters the expression of the *ACE* enzyme and may affect cardiovascular health. This polymorphism results from the insertion (I) or deletion (D) of a 287 bp Alu sequence within intron 16 of the *ACE* gene, resulting in three distinct genotypes: II, ID, and DD.[74] Increased activity of the ACE enzyme, often associated with the D allele, promotes the production of angiotensin II while decreasing the half-life of bradykinin. This dynamic may influence inflammatory responses following muscle injury, given that angiotensin II and bradykinin play significant roles in inflammation.[75] Sierra et al. have suggested a link between *ACE I/D* gene polymorphisms and susceptibility to postexercise muscle injuries, particularly sport-related muscle damage (SRMD). These polymorphisms could influence sensitivity to such damage by modulating the inflammatory response.[30] According to research, several studies have highlighted the association of the D allele of the *ACE I/D* polymorphism with a lower risk of muscle injuries and its protective role against these injuries.[29–31] Other studies have reported a correlation between the D allele and lower CK levels,[32] a marker of muscle damage, following intense efforts such as triathlons[55] and marathons.[30] Conversely, another study found that athletes with the ID genotype had a higher incidence of sports injuries compared to those with the DD and II genotypes.[76] Moreover, an interaction between polymorphisms was observed: the combination of the X allele of the *ACTN3* gene with the II genotype of *ACE I/D* resulted in a twofold increase in injury incidence per season.[28] Other studies found no association between the *ACE I/D* polymorphism and muscle injuries.[16,33,47,49,51,52] Overall, these findings highlight the complexity of *ACE* polymorphism effects, which may depend on genotype interactions, type of sport, and study population. Furthermore, the G allele of the *NOS3* gene (1799983) appears to have reduced sensitivity to proteolytic cleavage, which may promote increased enzyme activity and, consequently, enhanced production of *NOS3*.[77] To date, only two studies have investigated the association between *NOS3* gene polymorphisms and injuries in athletes. One study conducted with football players found that the rs1799983 polymorphism of the *NOS3* gene was associated with the occurrence of hamstring injuries.[47] In contrast, another study in high-level Brazilian athletes did not find any significant association between the rs1799983 polymorphism of the *NOS3* gene and MP or traumatic muscle

injuries.[58] These contrasting results suggest that *NOS3* variants may contribute to injury susceptibility in a context-specific manner, but further studies are needed to clarify their role.

Genes and hormonal regulation of muscle in athletes

Hormonal regulation, including estrogen, is crucial for stress response and muscle function. Imbalances can impair muscle function, increasing injury risk in athletes. Polymorphisms in the *ESR1* gene (rs2234693 C/T, rs9340799 G/A) affect receptor expression and estrogen action. The C allele of rs2234693 may protect against muscle injuries by reducing muscle stiffness.[42] Overall, *ESR1* variants may influence susceptibility to muscle injuries by affecting hormonal regulation of muscle tone and recovery. Genes, muscle growth, and development in athletes. Growth factors are essential in tissue regeneration and cellular functions, with genetic variants influencing injury prevention in athletes. Our research found that the *GEFT* gene, which encodes a guanine-nucleotide exchange factor activating Ras-related proteins Rac1 and Cdc42—key regulators of cell proliferation, migration, and cytoskeletal dynamics.[78] The rs11613457 polymorphism of the *GEFT* gene correlates with muscle injury and recovery time in football players, while the rs4227 polymorphism in the *SOX15* gene relates to injury rates.[43] However, two studies on elite football players found no significant association between rs4227 and injury severity or soft tissue injuries.[63,79] Genes encoding inhibitory proteins for connective tissue remodeling, like the *NOGGIN* gene, are crucial for maintaining balance. A study found a higher incidence of muscle injuries in carriers of the GG genotype (rs1372857), who was more likely to suffer moderate to severe injuries in elite Australian football players.[40] Variants regulating muscle growth, cytoskeletal dynamics, and connective tissue remodeling collectively contribute to individual differences in injury risk and recovery potential.

MicroRNAs associated with sports-related muscle injuries

Specific muscle miRNAs, or myomiRs, are emerging as potential biomarkers for assessing muscle damage in athletes. Studies show that plasma levels of miRNAs such as miR-1-3p, miR-133a-3p, miR-133b, miR-208a-3p, miR-208b-3p, and miR-499a-5p (Table 2) significantly increase after intense physical efforts, such as running, and are associated with muscle damage. A correlation between miRNA levels and decreased countermovement jump height 24 h after running suggests their link to muscle fatigue and the degree of muscle damage.[21] Additionally, endurance exercise increases miR-126 levels, while weight training elevates miR-133, indicating specific tissue damage profiles based on exercise type.[80] Research also highlights the role of circulating miRNAs in atrial remodeling in marathon runners. Specifically, increased plasma levels of miR-1, miR-30a, and miR-133a have been correlated with left atrial size, suggesting their potential as biomarkers for arrhythmia risk in endurance athletes.[19] Similarly, plasma levels of miR-133a and miR-206 have been shown to increase in circulating levels after a half-marathon, although no significant correlation was found with muscle stress

Table 2. MicroRNAs, their targets, and their role in muscle injuries.

MiRNA	Known target(s)	Role in healing/biomarker function
miR-1-3p	HDAC4	Regulates muscle growth and repair; marker of muscle damage after exercise.[21,83]
miR-133a-3p	SRF, IGF-1R, SERCA-2	Involved in myoblast differentiation; potential biomarker of muscle stress and atrial remodeling in endurance athletes.[83,84]
miR-133b	SRF, IGF-1R, SERCA-2	Promotes muscle regeneration; elevated after resistance training.[83,84]
miR-208a-3p	MYH6	Cardiac-specific; associated with cardiac adaptation and muscle damage following intense exercise.[21,85]
miR-208b-3p	MYH7, SOX6	Cardiac-specific; increases post-exercise, linked to skeletal and cardiac muscle stress.[21,85]
miR-499a-5p	MYH7B, SOX6	Involved in cardiac and skeletal muscle regulation; marker of exercise-induced muscle damage.[21,85]
miR-126	VEGF, SPRED1	Promotes angiogenesis and vascular repair; elevated after endurance exercise.[80,84]
miR-30a	Smarcd2, Snai2, Tnrc6a	Regulates autophagy and cardiac protection; correlated with left atrial enlargement in marathon runners.[84,86]
miR-206	Pax7, Ccnd1, IGF-1	Controls regeneration of muscle fibers; increases after half-marathon running.[81,84,87]
miR-31	Myf5, Dystrophin	Involved in satellite cell activation and muscle repair; decreased at 24 h post-exercise, indicating delayed recovery regulation.[82,87]

Ccnd1: cyclin D1; HDAC4: histone deacetylase 4; IGF-1R: insulin-like growth factor 1 receptor; Myf5: myogenic factor 5; MYH6: myosin heavy chain 6; MYH7: myosin heavy chain 7; MYH7B: myosin heavy chain 7B; Pax7: paired box 7; SERCA-2: sarcoplasmic/endoplasmic reticulum calcium ATPase 2; Smarcd2: SWI/SNF-related BAF chromatin remodeling complex subunit D2; Snai2: Snail family transcriptional repressor 2; SOX6: SRY-box transcription factor 6; SRF: serum response factor; SPRED1: Sprouty-related EVH1 domain-containing protein 1; Tnrc6a: trinucleotide repeat containing adaptor 6A; VEGF: vascular endothelial growth factor.

markers such as CK or troponin T.[81] Lastly, miRNAs encapsulated in extracellular vesicles (EVs) were measured in plasma at 0, 2, and 24 h after muscle-damaging exercise, revealing overall stability of myomiRs over this period, except for miR-31, whose expression significantly decreased at 24 h, suggesting delayed regulation linked to muscle recovery.[82] These findings suggest that circulating miRNAs, particularly those associated with EVs and plasma, could serve as sensitive, noninvasive biomarkers for early detection of muscle stress or injury. Monitoring miRNA levels through blood samples offers a practical and minimally invasive method compared to muscle biopsies. Furthermore, several miRNAs demonstrate early changes in expression, often within hours postexercise and show relative stability in circulation, making them suitable for real-time or short-term postexercise monitoring. Such biomarkers could be highly valuable for coaches, athletes, and medical staff in guiding training loads, identifying subclinical muscle damage, and preventing overuse injuries by allowing timely intervention before structural muscle damage occurs.

Table 3. Genetic variants associated with bone injuries in athletes.

Genes	Full name	Polymorphisms	Localization	References
CTR	Calcitonin receptor	rs1801197	7q21.3	89
LPR5	Receptor-related protein 5	rs2277268	11q13.2	89
MCT1	Membrane monocarboxylate transporter 1	rs1049434	1p13.2	51
P2RX7	P2X purinoceptor 7	rs3751143	12q24.31	90
RANK (TNFRSF11A)	TNF receptor superfamily member 11a	rs3018362	18q21.33	88
RANKL (TNFSF11)	TNF superfamily member 11	rs1021188	13q14.11	91
OPG (TNFRSF11B)	TNF receptor superfamily member 11b	rs4355801	8q24.12	88
SOST	Sclerostin	rs1877632	17q21.31	92
VDR	Vitamin D receptor	rs2228570 rs731236 rs10735810	12q13.11	14,92
Wnt16	Wnt family member 16	rs3801387	7q31.31	93

Genetic polymorphisms related to bone injuries in athletes

Bone injuries, including fractures, cracks, and contusions, result from trauma, repetitive activity, or excessive load.[88] Research shows significant links between genetic polymorphisms and susceptibility to bone injuries in athletes. Ten genes and 12 polymorphisms have been associated with these injuries (Table 3).

Genes, bone remodeling, and homeostasis in athletes

Bone homeostasis is primarily regulated by bone remodeling, a continuous process mediated by osteoclasts and osteoblasts. Other contributing factors include calcium and phosphate metabolism, hormonal signaling (e.g., PTH, vitamin D), and mechanical loading. The balance between bone formation and resorption is critical for preventing injuries and disorders such as osteoporosis. Several polymorphisms have been identified as influencing these regulatory pathways.[88,94] For example, the RANK/RANKL/OPG system has been linked to stress fracture risk. In elite athletes, the A allele of rs3018362 *(RANK)* and the AA genotype of rs1021188 *(RANKL)* were associated with higher susceptibility, whereas rs4355801 *(OPG)* showed no clear association, although carriers of its rare allele might experience multiple fractures.[88] Similarly, the T allele of rs9594738 *(RANKL)* correlated with reduced cortical cross-sectional area in adolescent football players, suggesting structural vulnerability.[91] Moreover, Genetic variations in *SOST* encoding sclerostin are associated with bone disorders such as osteoporosis and can affect bone mineral density (BMD), although associations with stress fractures are inconsistent.[95] The TC genotype (rs1877632) has been linked to stress fractures in elite athletes,[92] whereas other studies in football players and meta-analyses in active participants found no significant association.[91,94] In addition,

vitamin D receptor *(VDR)* polymorphisms have demonstrated a stronger and more consistent influence on bone health in athletes. Multiple SNPs (rs1544410, rs731236, rs7975232, rs10735810, and FokI rs2228570) indicate that specific genotypes modulate fracture risk and BMD. Notably, the TT and FF genotypes of rs731236 and rs10735810 are associated with an increased risk of fractures,[92] whereas the FF genotype of FokI in Italian athletes appears protective, correlating with higher total body mineral density and a lower prevalence of low back pain.[14] In football while in football the FF genotype corresponded to a 7.7% lower bone mineral content in the upper lumbar spine compared to FF genotype.[96] In Arab athletes, the CT and TT genotypes of the *VDR* rs10735810 were linked to a higher incidence of stress fracture injuries, with the TT genotype specifically associated with a greater risk of severe fractures compared to moderate or mild injuries.[97] These findings underscore the relevance of *VDR* polymorphisms for predicting bone fragility and tailoring training or nutritional interventions.

Genes, bone, and BMD in athletes

The differentiation of bone cells, osteoblasts, and osteoclasts is an important element in bone development and in the repair mechanism following a bone injury. Several genetic variants have been characterized due to their involvement in the different signaling pathways involved in the proliferation and differentiation of cells responsible for bone formation, as well as in maintaining the balance between bone formation and resorption.[89,98] A study conducted on young Finnish men eligible for mandatory military service at age 18 (lasting 6–12 months) investigated the impact of several *LRP5* gene polymorphisms, including rs2277268, under conditions of intense physical exertion. This basic training included walking, jogging, cycling, military drills, and combat exercises, imposing a significant physical load. The GA genotype of rs2277268 was significantly more frequent in individuals who developed stress fractures compared to those without such injuries. Furthermore, the A-G-G-C haplotype (comprising rs2277268, rs4988321, rs556442, and rs3736228) of the *LRP5* gene, when combined with the C-A haplotype of the *VDR* gene (rs10735810 and rs1544410), was associated with a nearly fourfold increase in femoral neck stress fracture risk.[89] In contrast, other studies, including a systematic review and a cohort study among Caucasian elite athletes, found no significant association between rs3736228 and stress fractures. These discrepancies may be due to differences in study populations (e.g., elite athletes vs. military conscripts), physical stress exposure, sample sizes, or fracture site definitions. Additionally, individual SNPs may have limited predictive power, whereas haplotype-based analyses may better capture the cumulative genetic risk in specific physiological contexts.[92,94] In fact, research has shown that variations in BMD are present between the runners and nonathletes with genotype AA carrying the same *WNT16* rs3801387 genotype, indicating a possible genetic interaction related to mechanical loading factors in endurance runners.[98] Furthermore, another study indicated that the frequency of the *WNT16* rs3801387 genotype is significantly different between athletes and controls, with the "at-risk" A allele being approximately 5% more frequent among athletes, suggesting a higher genetic susceptibility to bone injuries in elite endurance runners.[93] However, no association was

found in two studies between the rs3801387 polymorphism and stress fractures, as well as bone phenotypes in elite athletes and football players.[91,92] Overall, these findings suggest that while certain variants of *LRP5, VDR*, and *WNT16* may contribute to bone fragility under high mechanical loading, results remain inconsistent across studies. Haplotype-based analyses and gene–gene interactions seem more informative than single SNPs, but the variability between cohorts highlights the influence of training load, sport discipline, and population background. Overall, the evidence indicates that bone injury susceptibility in athletes is shaped by a polygenic architecture interacting with environmental and sport-specific factors, underscoring the need for integrative approaches rather than isolated genetic associations.

Genes, response to injury, and inflammation in athletes

The inflammatory response is a key step in healing after an injury, through the recruitment of immune cells to the injury site. Several genes are involved in regulating the inflammatory response by recruiting immune cells, such as macrophages and monocytes, which become activated by releasing pro-inflammatory cytokines and chemokines essential for healing.[47,91] For example, a study conducted on football players examined two polymorphisms of the *P2RX7* gene (*rs3751143* and *rs1718119*). However, no significant association was found. The C allele of *rs3751143* has been associated with reduced bone strength and functional loss of the receptor, whereas the T allele of *rs1718119* is linked to a 14% and 38% increase in bone density and cortical thickness at tibial sites. These findings highlight the essential role of *P2X7* in bone physiology while emphasizing the complexity of its relationship with bone injuries in athletes.[91] Additionally, the *P2X7R* variant rs3751143 (Glu496Ala – loss of function) is associated with an increased risk of stress fractures, whereas the rs1718119 variant (Ala348Thr – gain of function) is linked to a reduced risk in military conscripts and elite athletes ($p < 0.05$). These associations further support the role of genetic predisposition in the development of stress fractures.[90] These findings indicate that *P2RX7* gene variants modulate bone strength and fracture susceptibility, highlighting the importance of genetic predisposition in bone injury risk and the need to consider individual genetic profiles alongside training and sport-specific demands.

Genes, calcium, and mineral regulation in athletes

Calcium and minerals are crucial for bone function, influencing bone resorption and BMD. Polymorphisms in the *CTR* gene, such as Alu I, have shown variable associations with BMD and osteoporosis. For instance, the interaction between the *CTR* C allele and the VDR C-A haplotype has been linked to reduced parathyroid hormone production, offering protection against fractures in a Finnish cohort of 72 military conscripts with femoral neck stress fractures, compared to 120 healthy controls.[89] However, no significant association was found between the *CTR* rs1801197 polymorphism and stress fractures in a cohort of 518 British elite athletes.[92] Overall, *CTR* gene variants appear to influence bone metabolism and fracture risk in a context-dependent manner, with their

impact modulated by interactions with other genetic factors such as *VDR* haplotypes and population-specific characteristics.

Genes, metabolism, and transport in athletes

The *MCT1* gene encodes a transporter protein involved in lactate and pyruvate transport, crucial for energy metabolism and acid-base balance. A study found that the AA genotype of the rs1049434 polymorphism in *MCT1* is linked to a higher incidence of bone injuries in rugby players.[51] This evidence suggests that the *MCT1* rs1049434 polymorphism may serve as a potential genetic marker for bone injury susceptibility, emphasizing the relevance of energy metabolism pathways in predicting fracture risk among athletes.

MicroRNAs involved in bone injury and repair in athletes

In addition to genetic variants, epigenetic mechanisms, particularly miRNAs, have emerged as key regulators of gene expression in bone physiology and injury response. MicroRNAs are vital regulators of bone formation, remodeling, and regeneration, with significant implications for bone injuries in athletes. These small noncoding RNA molecules regulate gene expression posttranscriptionally, influencing the differentiation and function of bone cells, particularly osteoblasts. miR-21 plays a central role in osteoblast differentiation, affecting key factors such as Runx2, Osterix (OSX), and Mef2c. A deficiency in miR-21 reduces osteoblastic activity and increases sclerostin (SOST), an inhibitor of bone formation, which suggests its protective role in bone fragility (Table 4).[99] Other miRNAs, including miR-497-5p, miR-155-5p, miR-423-5p, and miR-365-3p, are associated with fracture healing, osteoporosis, and osteoarthritis. These miRNAs, detectable in blood, hold promise as noninvasive biomarkers for diagnosing and monitoring bone disorders.[100] Physical exercise also affects circulating miRNAs, which are linked to energy homeostasis and the cell cycle. This modulation may influence bone adaptation to mechanical stress, further emphasizing the role of miRNAs in bone health.[101] Therapeutic strategies using miRNA administration, such as miR-26a, have shown promise in promoting bone regeneration in animal models.[22] However, individual variability in miRNA expression, influenced by genetic polymorphisms in miRNA genes and epigenetic mechanisms such as DNA methylation and histone modifications, can modulate the posttranscriptional regulation of genes involved in bone formation and repair. This variability, therefore, contributes to individual differences in susceptibility to stress fractures or microstructural bone injuries in athletes.[102] Understanding these variations is crucial for developing personalized treatments. Further longitudinal studies are essential to explore miRNA regulation mechanisms and their potential for personalized medicine in sports injuries. These studies could lead to the development of targeted therapies for enhancing bone repair and preventing injuries, ultimately improving the recovery and performance of athletes.

Genetic polymorphisms related to concussions in athletes

Concussions, or mild traumatic brain injuries (mTBIs), are a significant concern in sports, especially among young athletes, where they result from rapid head movements causing

Table 4. MicroRNAs, their targets, and their role in bone injuries.

MiRNA	Known target(s)	Role in healing/biomarker function
miR-21	Runx2, Osterix (OSX), Mef2c, SOST	Promotes osteoblast differentiation; deficiency increases SOST, impairing bone formation; protective in bone fragility.[99]
miR-497-5p	CcnD2, CcnE1, Insr, Igf1r	Downregulated by IL-6; involved in cell cycle regulation and muscle regeneration pathways. Its inhibition increases Insr and Igf1r, key genes in the IGF signaling pathway supporting muscle hypertrophy and repair. Overexpression induces myotube atrophy, suggesting a role in negative feedback or compensatory regulation during cachexia.[103]
miR-155-5p	Aldh1 1, Nek2, Bub1b, Ramp3, Slc16a4, Plce1, Dync1i1, Nr1h3	Regulates genes involved in metabolism, cell cycle, and immune response; linked to muscle atrophy and dystrophies; promotes M2 macrophage infiltration.[104]
miR-423-5p	SRF	Negatively regulates SRF, leading to reduced expression of myogenin and myogenic markers; inhibits myogenic differentiation.[105]
miR-365-3p	RUNX2	Negatively regulates osteogenic differentiation of hBMSCs by targeting RUNX2; overexpression impairs mineralization and promotes osteoporosis progression.[106]
miR-26a	GSK3β, SMAD1	Promotes osteogenic differentiation and angiogenesis; enhances bone repair and strength when delivered via peptide nanoparticles in a hydrogel.[107]

Aldh1l: aldehyde dehydrogenase 1 family member L1; Bub1b: BUB1 mitotic checkpoint serine/threonine kinase B; CcnD2: cyclin D2; CcnE1: cyclin E1; Dync1i1: dynein cytoplasmic 1 intermediate chain 1; GSK3β: Glycogen synthase kinase 3 beta; hBMSCs: human bone marrow mesenchymal stem cells; Igf1r: insulin-like growth factor 1 receptor; IL-6: interleukin-6; Insr: insulin receptor; Mef2c: myocyte enhancer factor 2C; Nek2: NIMA-related kinase 2; Nr1h3: nuclear receptor subfamily 1 group H member 3; OSX: Osterix; Plce1: phospholipase C epsilon 1; Ramp3: receptor activity-modifying protein 3; RUNX2: Runt-related transcription factor 2; Runx2: Runt-related transcription factor 2; Slc16a4: solute carrier family 16 member 4; SMAD1: mothers against decapentaplegic homolog 1; SOST: sclerostin; SRF: serum response factor.

brain impact with the skull walls.[108] Previous research has shown associations between genetic variations and concussion susceptibility, identifying six genes with eight linked polymorphisms (Table 5).

Genes, neurodegeneration, and brain health in athletes

Even mild head trauma can lead to persistent neurological damage and promote neurodegenerative processes. Indeed, each impact disrupts neural structures and triggers an inflammatory response that, accumulated over time, increases the risk of neurodegenerative diseases and can impair cognition, memory, and mood.[108] Studies show contrasting

Table 5. Genetic variants associated with concussions in athletes.

Genes	Full name	Polymorphisms	Localization	References
APOE	Apolipoprotein E	rs429358 rs7412 rs405509 (−219 T/G)	19q13.32	109
COMT	Catechol-*O*-methyltransferase	rs4680	22q11.21	110
DRD4	Dopamine receptor D4	rs1800955 (−521C/T)	11p15.5	111
MAPT	Microtubule-associated protein tau	rs10445337	17q21.31	112
SLC6A4 (5-HTTLPR)	Solute carrier family 6 member 4	rs25531	17q11.2	110
SLC17A7	Solute carrier family 17 member 7	rs74174284	19q13.33	113

results regarding the association between *APOE* gene polymorphisms and the risk of concussions in athletes. Collegiate athletes carrying the TT genotype of the *APOE* G-219 T promoter have nearly 3 times the risk of having a history of concussions.[114] Conversely, protective associations have also been described, such as the *APOE ϵ4* allele linked to a 40% reduction in concussion risk[115] and the rs405509 TT genotype, which was overrepresented in nonconcussed controls compared to concussion cases (29% vs. 18%, $p = 0.043$) and associated with faster recovery after a concussion.[116] In addition, rare alleles of the E2, E4, and G-219 T promoter combination were significantly associated with concussion history.[109] However, not all studies confirmed these associations. In particular, several investigations found no significant link between *APOE ϵ4*, the *APOE G-219 T* promoter, or the tau polymorphisms Ser53Pro and His47Tyr and concussion risk.[13,117] These data indicate that *APOE* variants may influence vulnerability to concussion-related neurodegeneration, although population- and sport-specific factors appear to modulate these effects, highlighting the complexity of genetic contributions to brain injury outcomes.

Genes, neurotransmitter regulation, and cognitive function in athletes

The regulation of neurotransmitters and cognitive function is important element for the proper functioning of the brain and emotional balance. Neurotransmitters, such as dopamine, serotonin, acetylcholine, and glutamate, act as chemical messengers between neurons, influencing various aspects of cognition, including memory, attention, motivation, and decision-making. Numerous genes work to regulate these neurotransmitters, as an overproduction or deficiency can impair cognitive functions and contribute to the development of disorders such as depression, anxiety, and neurodegenerative diseases athletes.[118,119] The *COMT* rs4680 polymorphism provides an illustrative example. The Val/Val (G/G) genotype of the *COMT* rs4680 polymorphism, which is more prevalent among controls[110]and elite rugby athletes,[115] suggesting a potential role in cognitive performance, whereas study in football, soccer, baseball, softball, and basketball athletes

reported no association with concussion history,[117] and low-expression *5-HTTLPR (SLC6A4* rs25531) alleles, are underrepresented particularly among junior athletes, and are linked to reduced anxiety traits, indicating that personality characteristics may influence concussion risk in rugby,[110] with this polymorphism further linked to differences in psychological traits such as novelty seeking, self-management, and self-transcendence and to a higher frequency of the G allele and GA genotype in combat sports athletes compared to controls.[120] However, dopamine receptor genes *DRD2* and *DRD4* also contribute, with the *DRD4* CC genotype and the inferred *DRD2* (rs12364283–rs1076560)–*DRD4* (rs1800955) A–C–C allele combination associated with decreased concussion risk in rugby players personality traits,[118] the DRD4-521CC genotype correlating with sport-specific sensation-seeking in skiers without affecting impulsive sensation-seeking,[111] and *DRD2* rs1799732 is linked to reward dependence traits in MMA athletes,[121] whereas rs1800497 shows no significant association with concussion in collegiate athletes.[122] Polymorphisms affecting neurotransmitter systems (*COMT, 5-HTTLPR, DRD2/DRD4*) seem to shape cognitive performance, emotional regulation, and psychological traits in athletes, which in turn may alter susceptibility to concussion or impact recovery trajectories.

Genes, cytoskeletal stability, and neuroprotection in athletes

Cytoskeletal stability and neuroprotection are vital after concussions. An epistasis analysis showed a *COMT* (rs4680) and *MAPT* (rs10445337) G-C allele interaction, more common in elite rugby athletes, possibly affecting stress response and concussion risk.[112] However, *MAPT* Ser53Pro, His47Tyr,[114] and H1H1 genotypes showed no significant associations with athletes or cortical grey matter.[123] These findings suggest that such genetic interactions may modulate neuronal resilience and postconcussion recovery, highlighting that specific genetic profiles influence neuroprotective mechanisms rather than solely the likelihood of injury occurrence.

Genes and neurotransmitter transport in athletes

Neurotransmitter transporters are crucial for brain function, especially after concussions. Traumatic brain injuries can disrupt neurotransmitter balance, causing excitotoxicity and inflammation. The SNP rs74174284 in the *SLC17A7* gene promoter has been linked to concussion severity and recovery time, with carriers of the G allele showing longer recovery and reduced motor speed on the ImPACT test,[113] highlighting the importance of neurochemical homeostasis for postinjury brain adaptation.

MicroRNAs and sport-related concussions

Salivary and serum miRNAs are emerging as effective tools for diagnosing and monitoring concussions in athletes. Studies show that miRNAs, particularly the miR-27a-5p/miR-30a-3p ratio, can reliably differentiate concussed athletes from nonconcussed individuals, regardless of physical activity or contact sports.[124] Other research has identified specific serum miRNA signatures that can distinguish concussed athletes, including miRNAs such as let-7c-5p,

Table 6. MicroRNAs, their targets, and their role in brain injuries.

MiRNA	Known target(s)	Role in healing/biomarker function
let-7a-5p	TGF-B3, HMGA2	Involved in innate immune response in the brain; elevated post-concussion.[135]
let-7c-5p	Casp 3, HMGA2, IL-6	Modulates inflammation; proposed biomarker for mild traumatic brain injury (mTBI).[136]
let-7f-5p	NA	Part of SCRUM salivary signature; useful for rapid post-concussion screening (within 48 h).[132]
let-7i-5p	UCHL-1, IL-1β, IL-6, NMDA, IL-8	Neuroinflammation, neuronal damage, and excitotoxicity, thereby influencing recovery after concussion.[137]
miR-107	BACE1, GSK3B, GRN	Altered in neurodegeneration; affects neuronal plasticity.[138]
miR-122-5p	PLA2-IIA	Promising candidate for distinguishing mild TBI from controls.[139,140]
miR-135b-5p	NA	Neuroprotective effect by reducing neuronal apoptosis and oxidative stress.[141]
miR-142-3p	PICALM	regulating cellular processes modulating.[142]
miR-143-3p	ERK5, Bax and Casp 3	Included in SCRUM biomarker panel for salivary detection of SRC.[143]
miR-146a-5p	CFH, IRAK1, TSPAN12	Regulates neuroinflammation via NF-κB signaling.[144]
miR-151-5p	SCN4A	Involved in neuronal signaling and injury response, with its expression correlating to brain injury severity and neural tissue damage.[141]
miR-153-3p	APP,APLP2, NFe212	Neuroprotective; increased in the acute phase after concussion.[144]
miR-16	BCL2	Altered after concussion; regulates apoptosis and inflammation.[145]
miR-181c-5p	Cpne2, Gria1,Gria2	Responds to oxidative stress and inflammation post-injury.[146]
miR-18a-5p	NLRP3, ASC, casp 1	Modulates neuroinflammation by regulating inflammasome activation, thereby influencing inflammatory responses and neuronal damage after brain injury.[147]
miR-195	GABA	Associated with preseason cognitive scores; involved in synaptic plasticity.[141]
miR-20a	cyclin D1, E2F1, Ngn1	Regulates neuronal apoptosis, inflammation, and regeneration by targeting key cell cycle and differentiation proteins.[140]
miR-20a-5p	E2F1, TGFβR2	Regulates neuronal proliferation and apoptosis; altered post-repetitive head impacts.[141]
miR-221-3p	CDKN1B	Downregulated after concussion; inversely correlates with symptom severity.[148]
miR-223-3p	NLRP3	Regulates inflammasome activity; increases after brain trauma.[149]
miR-27a-3p	BMi1, Sox11, Zfp90, Mex3a	Reduced post-injury; modulates neuroinflammation.[140,150]

(continued)

Table 6. Continued.

MiRNA	Known target(s)	Role in healing/biomarker function
miR-27a-5p	FOXO1, PPARγ, IGF1R	Involved in neuroinflammation; miR-27a-5p/miR-30a-3p ratio discriminates concussed individuals.[124]
miR-28-3p	NA	Linked to inflammation and cellular stress; associated with repetitive brain injury.[126]
miR-30a-3p	NA	Regulates neuronal autophagy and neuroprotection; included in diagnostic miRNA ratios.[124]
miR-339-5p	APP, BACE1	Implicated in amyloid metabolism; potential marker of cumulative effects of head trauma.[151]
miR-362-3p	SCN4A	Involved in neuronal development and signal propagation; correlates with balance scores and may serve as a potential biomarker for neurological assessment.[141]
miR-425-5p	PTEN, BCL2L1	Downregulated in acute phase of SRC, returns to baseline later.[152]
miR-505	NA	Modulates neuroinflammation, cell survival, and apoptosis during brain injury response.[141]
miR-93-5p	NA	Plays a role in neuroprotection and tissue repair by regulating inflammatory and apoptotic pathways.[128]
miR-9-3p	Dmd, SAP97	Regulates synaptic plasticity and memory.[153]

APP: amyloid precursor protein; APLP2: amyloid beta precursor-like protein 2; ASC: apoptosis-associated speck-like protein containing a CARD; BACE1: beta-site APP-cleaving enzyme 1; Bax: BCL2 associated X, apoptosis regulator; BCL2: BCL2 apoptosis regulator; BCL2L1: BCL2 apoptosis regulator like 1; BMi1: B lymphoma Mo-MLV insertion region 1 homolog; CASP3: caspase 3; Casp 1: caspase 1; CFH: complement factor H; CDKN1B: cyclin-dependent kinase inhibitor 1B; Cpne2: Copine 2; Dmd: dystrophin; ERK5: extracellular signal-regulated kinase 5; E2F1: E2F transcription factor 1; FOXO1: Forkhead box O1; GABA: gamma-aminobutyric acid; GRN: granulin; Gria1: glutamate ionotropic receptor AMPA type subunit 1; Gria2: glutamate ionotropic receptor AMPA type subunit 2; GSK3B: glycogen synthase kinase 3 beta; HMGA2: high mobility group AT-Hook 2; HMGA2: high mobility group AT-Hook 2; IGF1R: insulin-like growth factor 1 receptor; IL-1β: interleukin-1 beta; IL-6: interleukin-6; IL-8: interleukin-8; IRAK1: interleukin 1 receptor associated kinase 1; Mex3a: Mex-3 RNA binding family member A; NA: not available; NFe212: nuclear factor erythroid 2–related factor 2 (likely NFE2L2); Ngn1: neurogenin 1; NLRP3: NLR family pyrin domain containing 3; NMDA: *N*-methyl-D-aspartate receptor; PLA2G-IIA: phospholipase A2 group IIA; PICALM: phosphatidylinositol binding clathrin assembly protein; PPARγ: peroxisome proliferator-activated receptor gamma; PTEN: phosphatase and tensin homolog; SAP97: synapse-associated protein 97; SCN4A: sodium voltage-gated channel alpha subunit 4; Sox11: SRY-box transcription factor 11; TGFβR2: transforming growth factor beta receptor 2; TGF-β3: transforming growth factor beta 3; TSPAN12: tetraspanin 12; UCHL-1: ubiquitin carboxy-terminal hydrolase L1; Zfp90: zinc finger protein 90.

miR-16-5p, miR-181c-5p, and miR-146a-5p. Some of these miRNAs display persistent alterations, detectable up to four months after the injury.[125] MiR-28-3p and miR-339-5p were also found to be linked to recurrent concussions, suggesting a correlation with the cumulative effects of trauma.[126] A study of 251 concussed athletes stratified by 22 salivary miRNAs identified ten distinct biological groups, with significant differences in concussion history, symptoms, and physiological pathways, which could contribute to more personalized approaches for concussion management.[127] Furthermore, miRNAs such as miR-16-5p, miR-18a-5p, miR-20a-5p, miR-93-5p, and miR-107 were modulated by accidental head impacts, repeated head impacts, or intense exercise, indicating their sensitivity to

neurophysiological stress.[128] Among university athletes, correlations between certain miRNAs (miR-195, miR-20a, miR-151-5p, miR-505, miR-9-3p, miR-362-3p) (Table 6) and preseason cognitive or balance scores were found.[20] In a study of Australian football players, decreases in plasma levels of miR-221-3p and miR-27a-3p after a sports-related concussion (SRC) were inversely correlated with symptom severity.[129] Several miRNAs, including miR-27b-3p, let-7i-5p, miR-142-3p, miR-107, and miR-135b-5p, were significantly upregulated in concussed athletes.[130] Acute increases in miR-153-3p, miR-223-3p, and miR-let-7a-5p were also observed in other cohorts, although some subjects showed no variation.[131] the Study of Concussion in Rugby Union through MicroRNAs identified a signature of 14 small noncoding salivary RNAs, including let-7f-5p, miR-143-3p, and RNU6-7, that enabled accurate diagnosis up to 48 h postmatch, highlighting the potential for timely diagnosis.[132] Additionally, subexpression of miR-425-5p was detected in the early phase following a SRC, with levels returning to normal shortly after.[133] Finally, a review comparing serum miRNAs to traditional clinical tests for mTBIs identified several relevant candidates, including miR-181c-5p, miR-195, and miR-122-5p.[134] These findings suggest that miRNAs represent a promising noninvasive and objective tool for the diagnosis and prognosis of concussions in athletes.

Ethical and translational considerations

The integration of genetic data in sports raises significant ethical and practical concerns that must be carefully addressed to ensure responsible and fair applications. Key issues include genetic privacy, necessitating strict protocols to protect athletes confidentiality and informed consent prior to any genetic testing or data use.[154,155] The question of fairness in sport emerges strongly, as genetic testing for talent identification risks genetic discrimination and inequity, potentially favoring athletes with access to advanced technologies while undermining equal opportunities for all participants. Furthermore, the ethical use of genetic information must consider the implications of genetic enhancements and testing on the integrity and spirit of competition, including the prevention of genetic doping and safeguarding athlete well-being.[155,156] These challenges emphasize the need for robust legal frameworks, universal guidelines, and ongoing ethical scrutiny to balance scientific progress with respect for human dignity, fairness, and the protection of athletes rights in the competitive environment.[157,158]

Methodological limitations of genetic association studies

Genetic association studies investigating the relationship between polymorphisms and susceptibility to sports-related injuries present several well-recognized methodological limitations that must be considered for a rigorous interpretation of findings. First, the small sample sizes of many studies reduce statistical power, increasing the risk of false-positive associations or inflated effect estimates, particularly when the variants investigated exert only modest effects on complex phenotypes.[159] Second, replicating results remains challenging due to heterogeneity across studies, including differences in study design, operational definitions of injury phenotypes, and environmental exposures.

Third, population stratification, systematic differences in genetic ancestry between cases and controls, can lead to spurious associations if not adequately addressed. Additional factors, such as publication bias and the lack of standardized tools for methodological quality assessment, further contribute to variability and poor reproducibility of results.[160] To enhance the robustness of evidence, it is recommended to assess genetic findings according to frameworks that integrate sample size, independent replication, rigorous control for confounding, and methodological strength, such as the human genetic evidence framework or conventional hierarchies of evidence.[161–163] Explicitly acknowledging these limitations is essential to contextualize the reliability of genetic associations and to guide the development of high-quality future research in sports genomics.

Conclusion and perspectives

This review synthesizes current evidence on associations between genetic and epigenetic factors and susceptibility to sports-related muscle damage, bone stress fractures, and concussions. Among the studies reviewed, 90 specifically investigated genetic polymorphisms, identifying the most consistently reported markers such as *ACTN3* R577X, *ACE* I/D, *AMPD1* rs17602729, *MCT1* rs1049434, and *MLCK* variants for muscle damage; *LRP5* haplotypes, *VDR* rs2228570/rs731236, *SOST* rs1877632, and *P2RX7* rs3751143 for bone injuries; and *APOE* promoter variants, *DRD4* rs1800955, and *COMT* rs4680 for concussion. Emerging miRNA candidates (e.g., miR-133a-3p, miR-208a-3p, miR-21, miR-28-3p, miR-339-5p) show promise as minimally invasive biomarkers for early detection and monitoring. Future research should prioritize large, sport-specific longitudinal cohorts integrating multiomics, functional validation, and ethical considerations to enable personalized injury prevention and recovery strategies in athletes.

Acknowledgments

This study was supported by the Mohammed VI Foundation for Sciences and Health.

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Authors' contributions

Amina Marrouh conceptualized and developed the study, conducted the literature review, authored the manuscript, and contributed to data analysis and interpretation. Arij El Haddouchi assisted with the literature review and reviewed and revised the manuscript. Souad Kartti critically reviewed the manuscript for intellectual content and approved the final version. Christophe Baudot and Younes Chagar reviewed the paper based on their expertise in the field of sports medicine. Elmostafa El Fahime, Saber Boutayeb, and Lahcen Belyamani supervised, revised, and validated the last draft. Taoufiq Dakka and Rachid Eljaoudi supervised, validated, and contributed to the writing, review, and editing process. All authors read and approved the final version of the manuscript and agree with the order of presentation of the authors.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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